Cellular Expert

8. RL Prediction

2025

1. Objective

This tutorial will show you how to add new MW links, manage and do predictions.

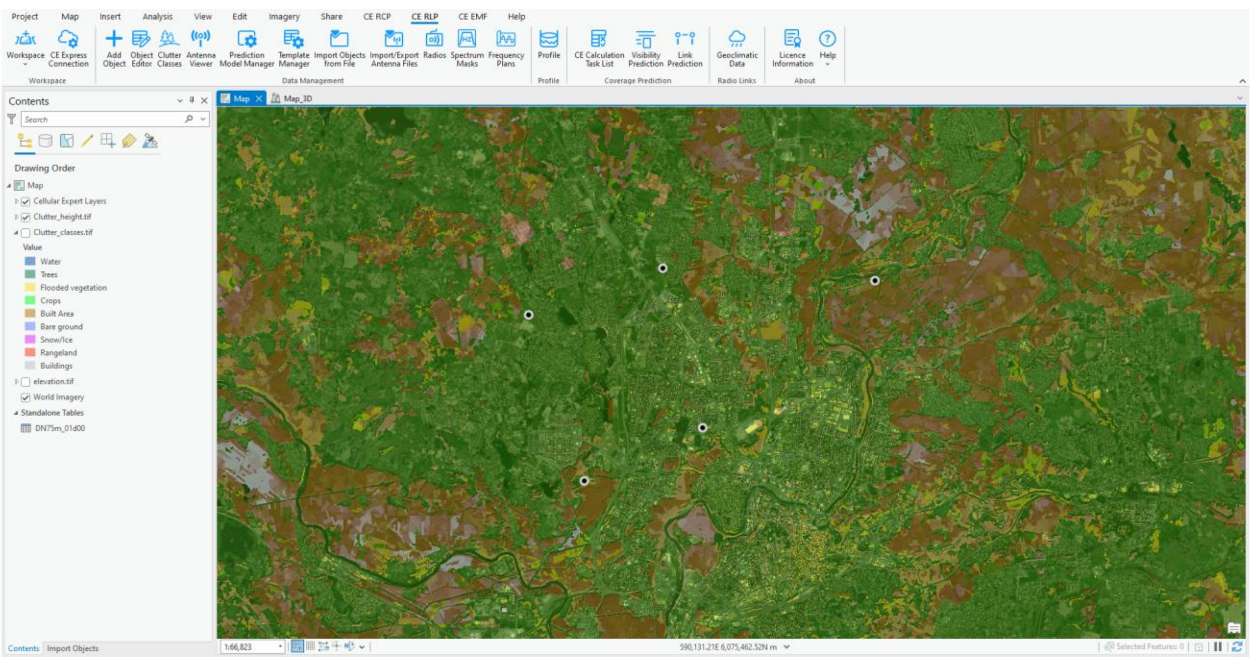
At the end of the exercise you will be able to:

- Create and import MW equipment.
- Create MW links in the project.
- Do MW Predictions.

2. Initial data

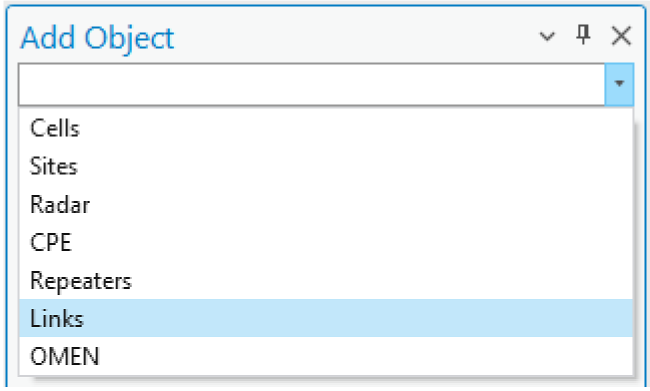
Prepared project with:

- Network objects.
- Geodata.
- Equipment and models.



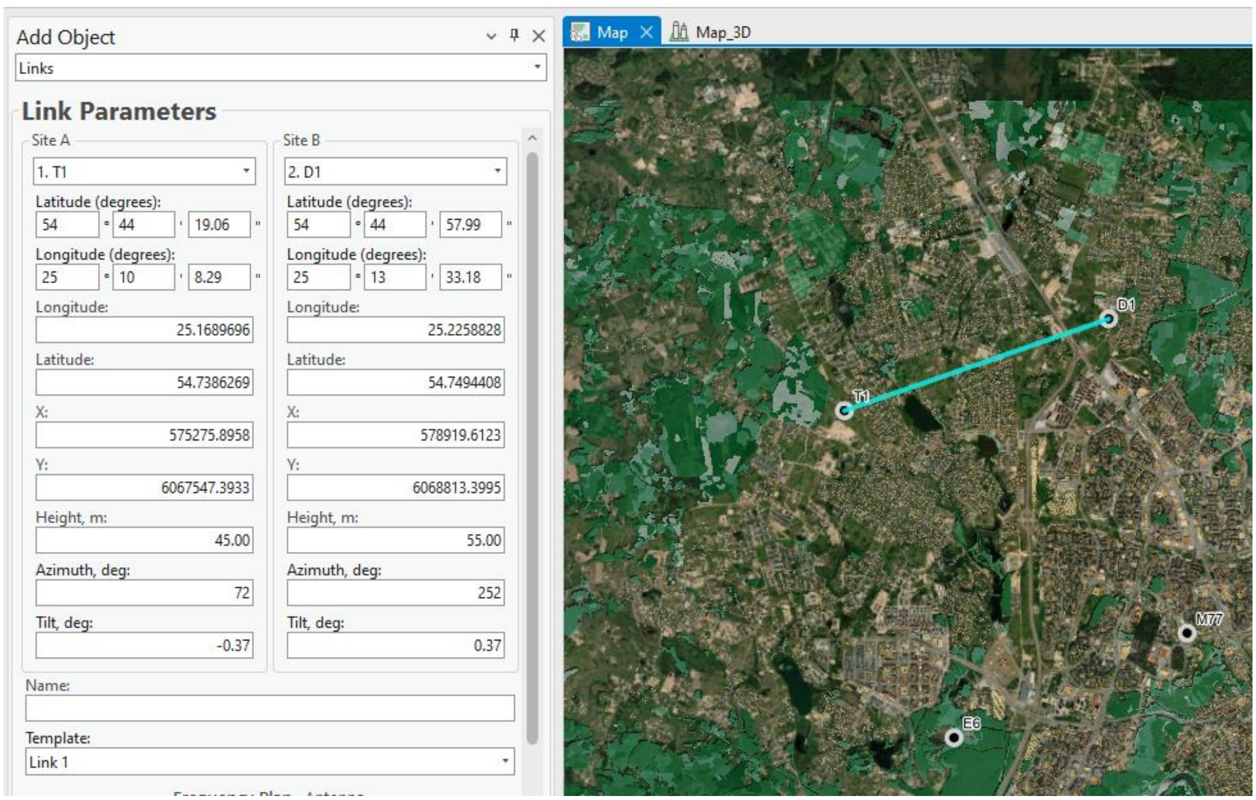
3. Add Links

The link objects can be imported or created manually using Add functionality. Open Add Object tool and select Link option from drop-down menu list.



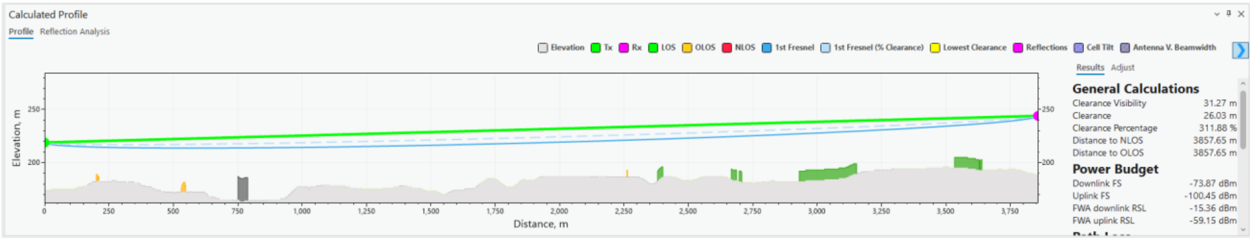
Click once on T1 site and second time to D1 site.

It will fill general parameters automatically.



Height, azimuth and other parameters are calculated automatically, leave them as it is. Press Show Profile button to preview topographical data between Site A and Site B, if there is LOS, what Fresnel zone percentage value, etc.

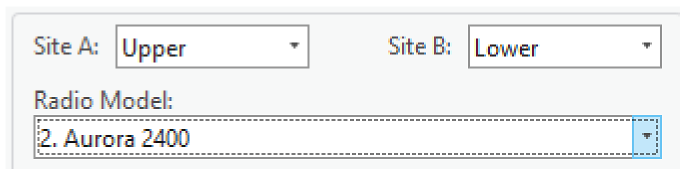
Profile will be generated.



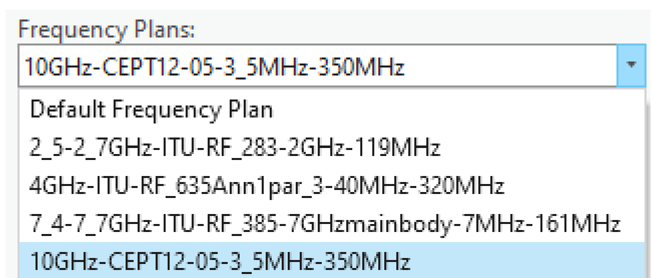
Close Profile windows.

Define parameters are defined below:

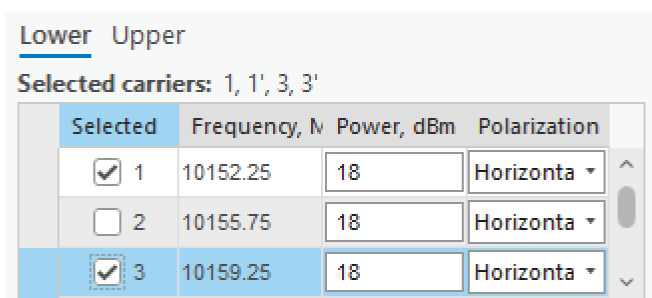
- Name: MW001
- Radio Model: 2. Aurora 2400



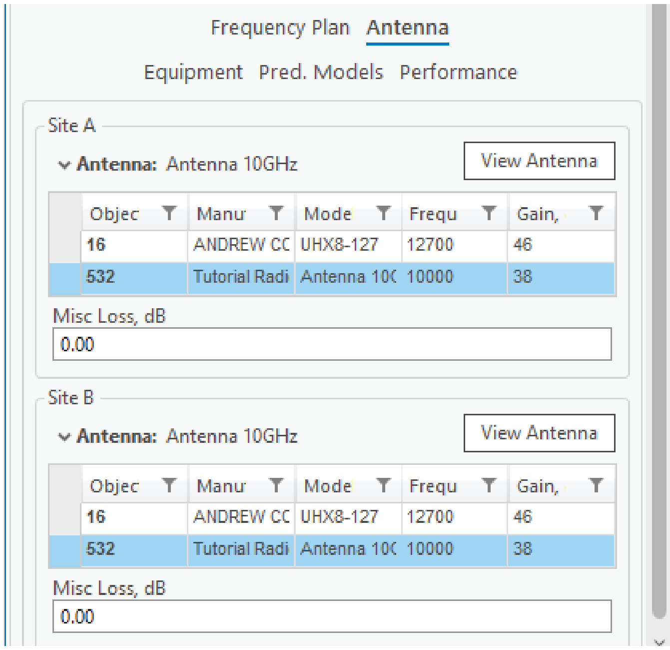
- Frequency Plan: 10GHz-CEPT12-05-3_5MHz-350MHz



- Carriers: 1 and 3 (1’ and 3’ will be automatically assigned).



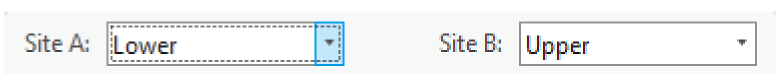
- Go to Antenna section and change antenna to Antenna 10MHz



Press Save Changes button.

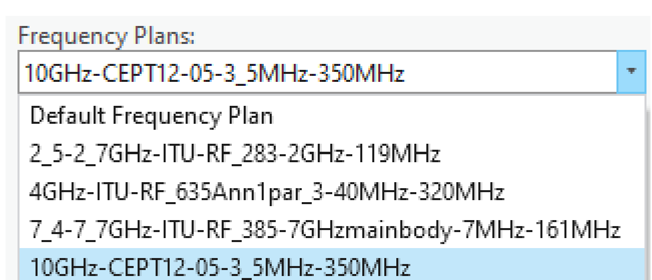
Do not close the dialog, create new links and define parameters:

- Between D1 and E2
    - Name: MW002
    - SiteA: Lower

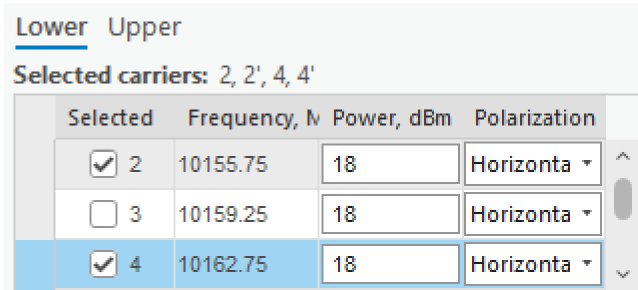


    - Radio Model: 2. Aurora 2400

    - Frequency Plan: 10GHz-CEPT12-05-3_5MHz-350MHz



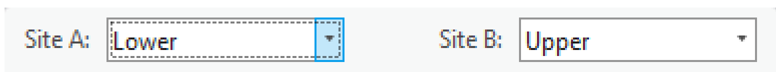
- Carriers: 2 and 4



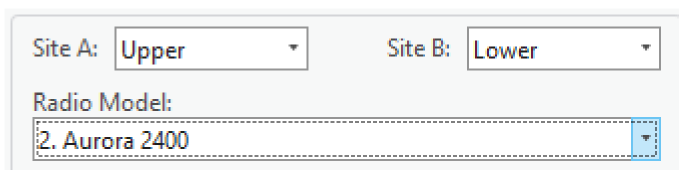
- Antenna for Site A and Site B: Antenna 10MHz

Press Save Changes

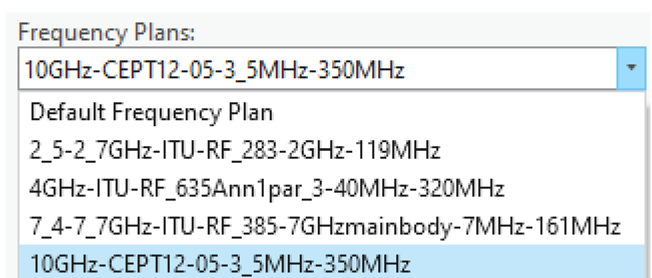
- Between E6 and M77
  - Name: MW003
  - SiteA: Lower



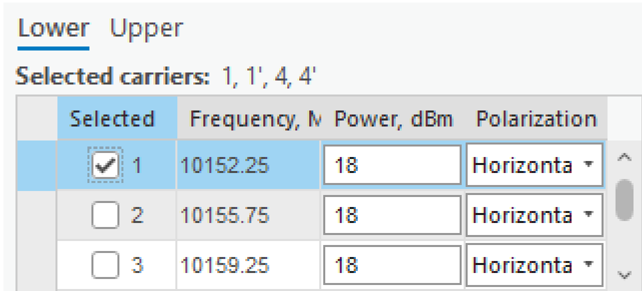
  - Radio Model: 2. Aurora 2400



  - Frequency Plan: 10GHz-CEPT12-05-3_5MHz-350MHz

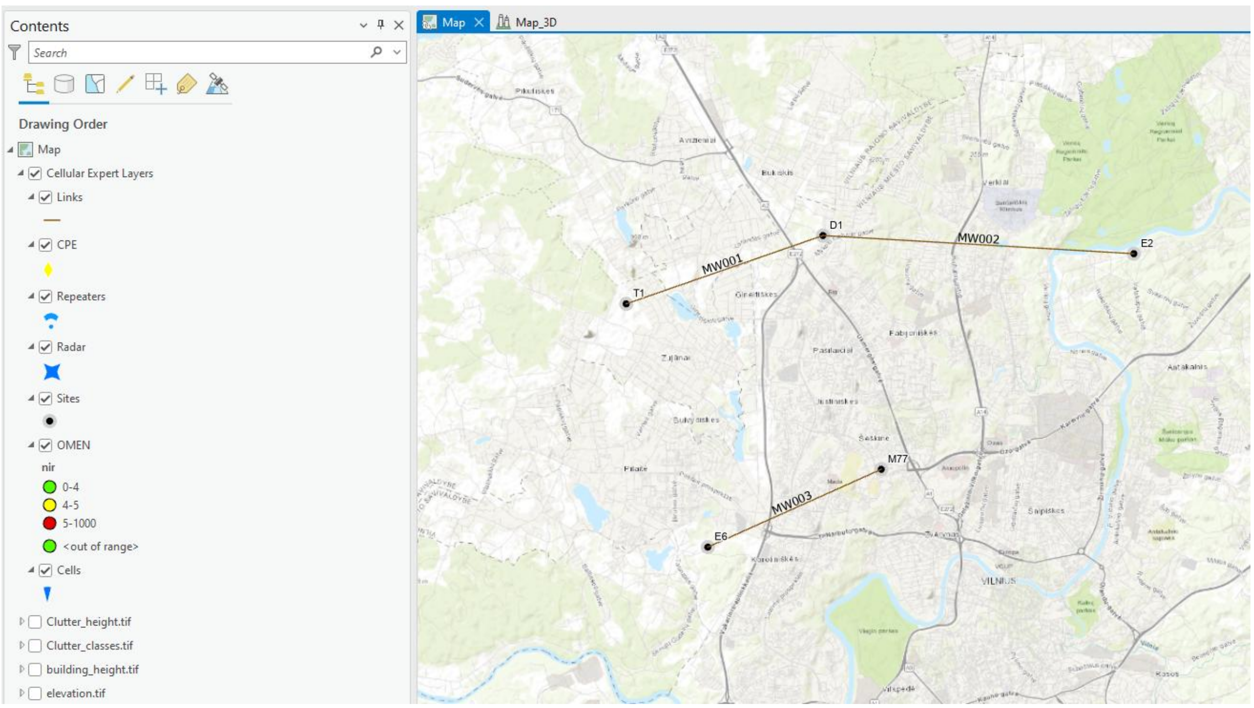


  - Carriers: 1 and 4



- Antenna for Site A and Site B: Antenna 10MHz

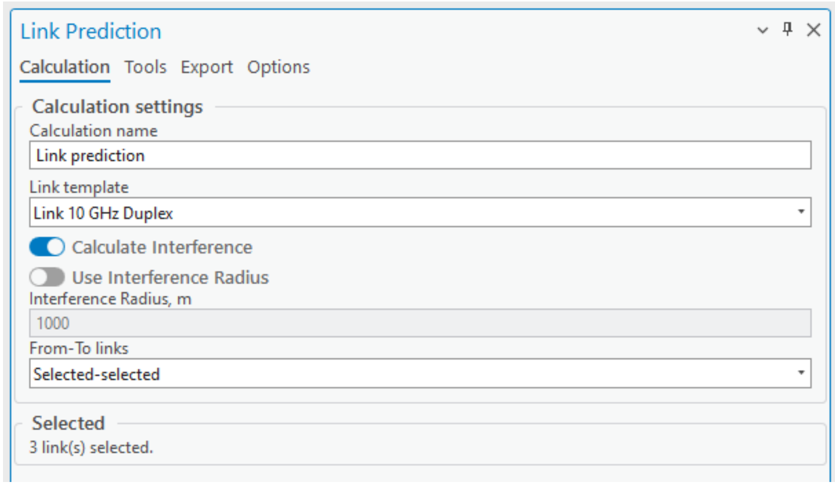
Press Save Changes



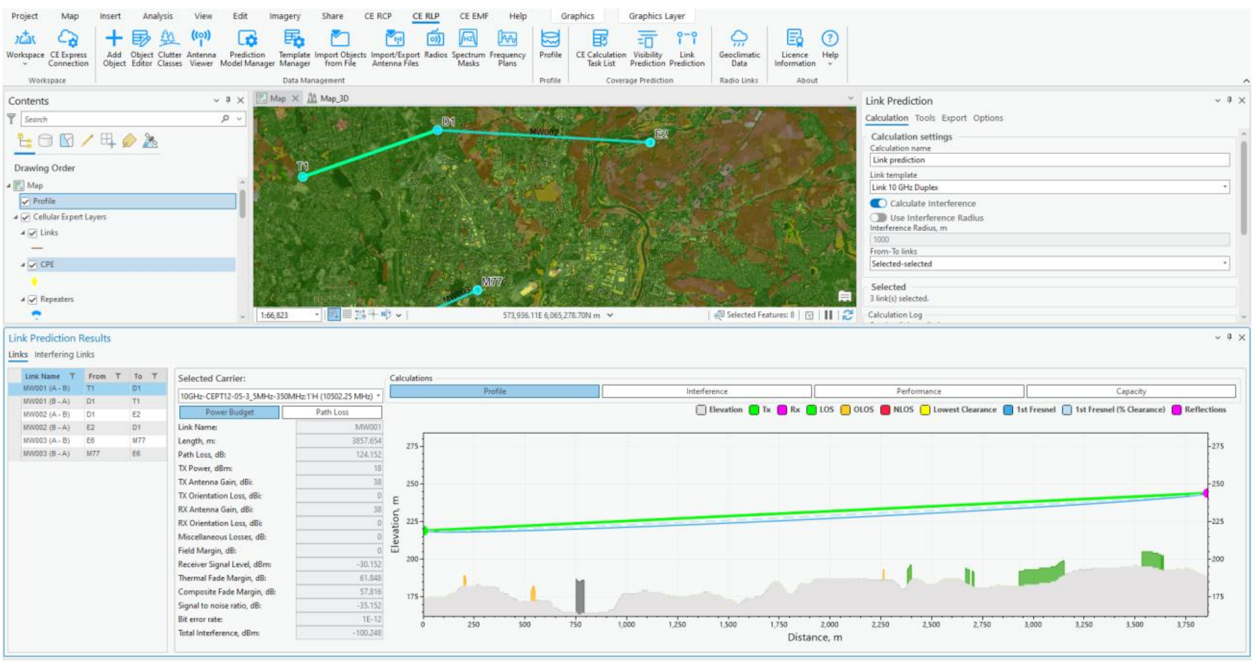
4. RL Predictions

The predictions are working between selected links and provides all necessary information about power budget, interference or availability.

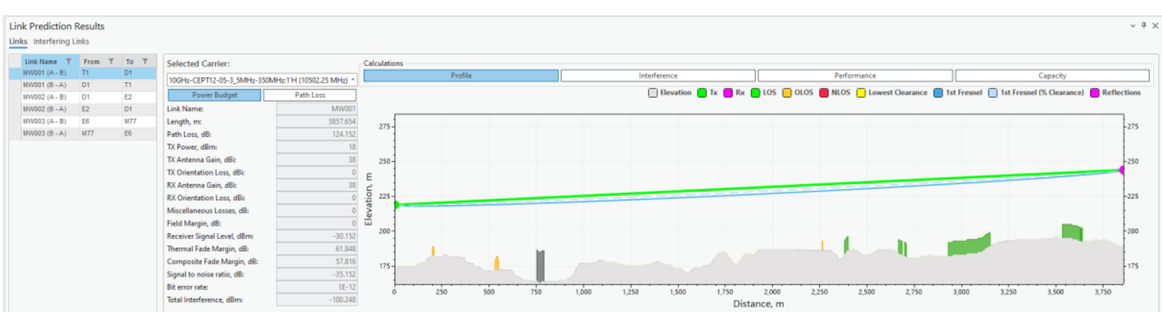
Select all links on the map and open Link Prediction tool. Before launching the predictions, enable Calculate Interference option.



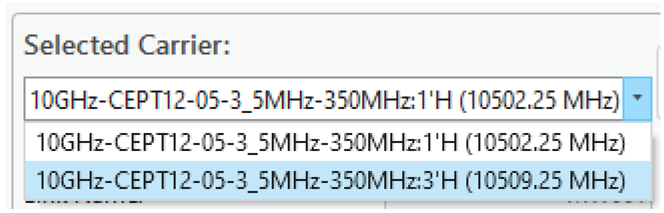
Press Run button and wait will prediction are completed and results loaded into your project.



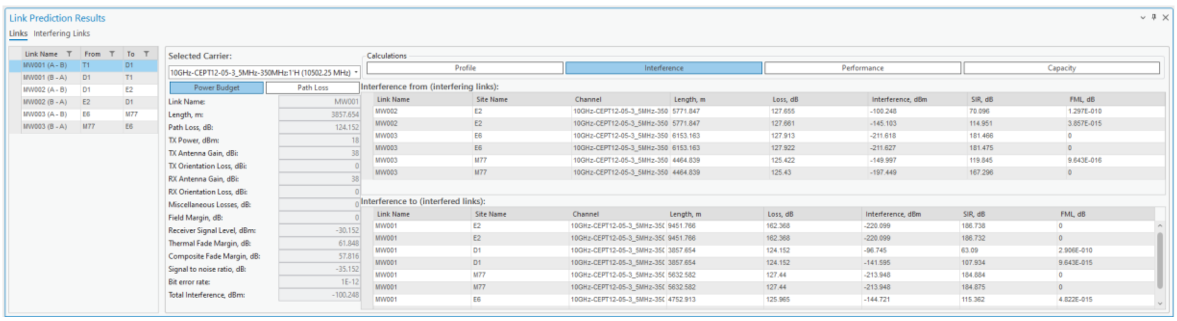
The primary results are Power Budget and Profile between Tx and Rx. On the left, select another link and results will be updated accordingly.



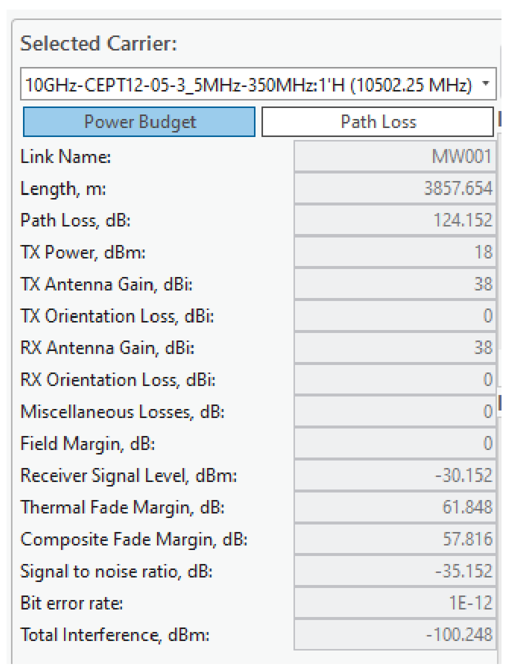
Selected Link can have several carriers, to preview the results expand Carrier option.



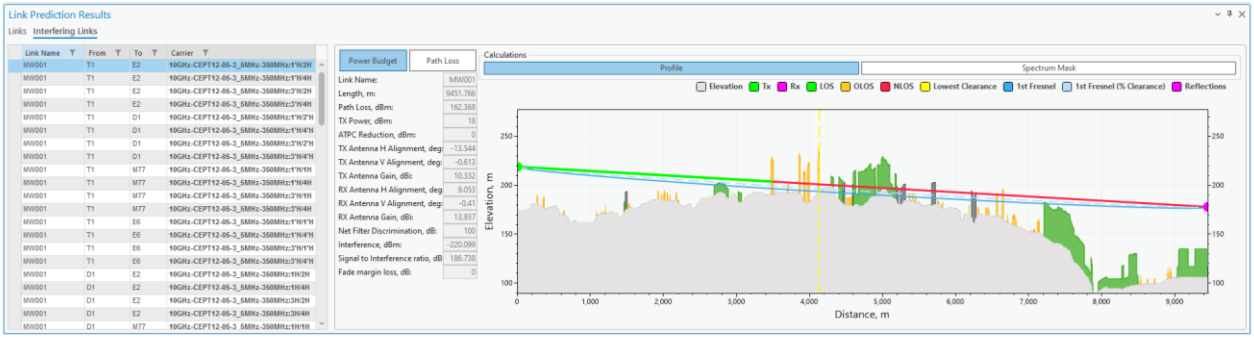
Click on Interference tab to display interfering links and interfered links information.



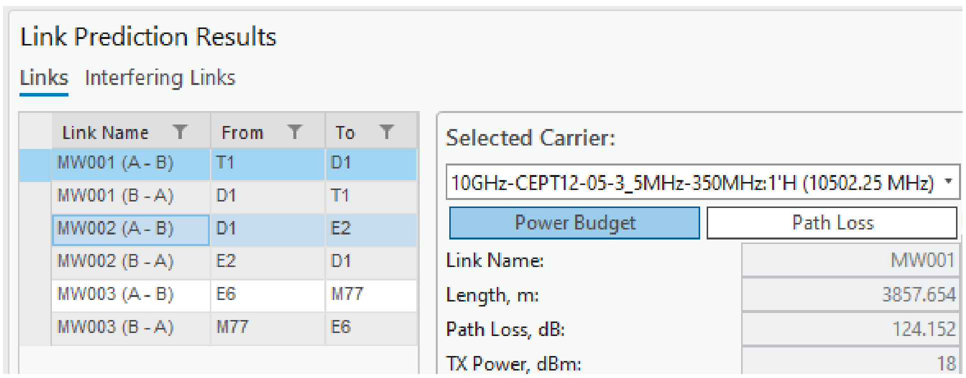
Total Interference is shown in Power Budget section.



Interference link predictions can be previewed too. Click on Interfering Link in top left corner and it will display selected interfering link Power Budget, Path Loss and draw a line on the map.



Go back to Links section, select MW001 from T1 to D1, Carrier 1'H.



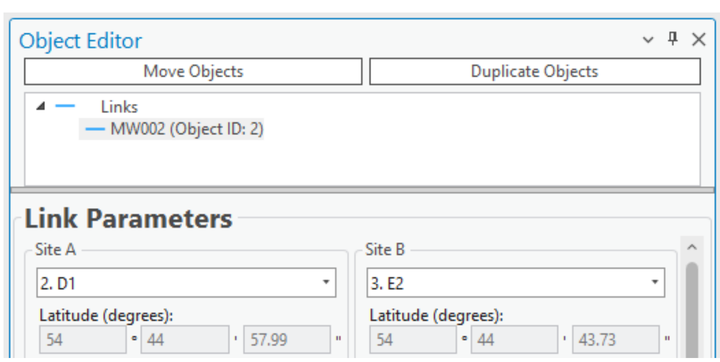
Click on Interference tab, and analize Inteference From table. Link MW002, from Site: E2 with Carrier 2'H interferes:

- Interference, dBm: -100.248 dBm
- FML, dB: 1.29E-10

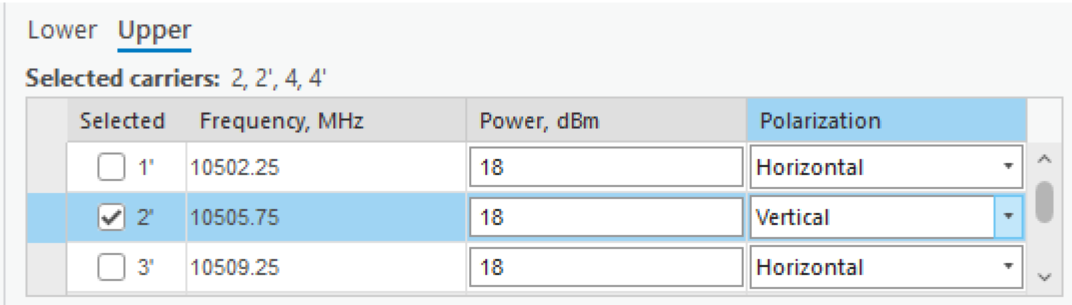
Interference from (interfering links):

Link Name	Site Name	Channel	Length, m	Loss, dB	Interference, dBm	SIR, dB	FML, dB
MW002	E2	10GHz-CEPT12-05-3_5MHz-350MHz:2'H	5771.847	127.655	-100.248	70.096	1.297E-010

Close Link Prediction results, select MW002 link on the map and open Object Editor. Double click on the link.



Find Frequency Plan table, select Upper and change carrier 2’ polarization to Vertical.



Lower Upper

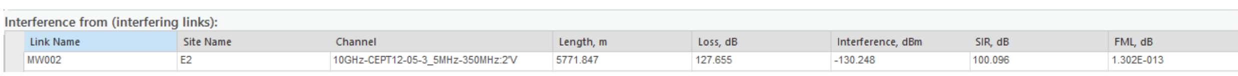
Selected carriers: 2, 2', 4, 4'

Selected	Frequency, MHz	Power, dBm	Polarization
☐ 1'	10502.25	18	Horizontal
☑ 2'	10505.75	18	Vertical
☐ 3'	10509.25	18	Horizontal

Save Changes.

Select all links on the map, open Link Prediction tool again and run predictions.

After predictions are done, open MW001 (A-B) interference predictions and preview how Interference From is changed for MW002 link, Site E2.



Interference from (interfering links):

Link Name	Site Name	Channel	Length, m	Loss, dB	Interference, dBm	SIR, dB	FML, dB
MW002	E2	10GHz-CEPT12-05-3_5MHz-350MHz:2'V	5771.847	127.655	-130.248	100.096	1.302E-013

Close Link Prediction Results dialog.

4.1 Geoclimatic data

Geoclimatic data, comprising factors such as rain rate, temperature, gaseous absorption, and multipath fading, significantly shapes the planning and efficacy of microwave links. Rain rate introduces the challenge of rain fade, where increased rainfall leads to greater signal attenuation. To address this, microwave link planners employ rain attenuation models, adjusting transmitter power or incorporating diversity techniques to maintain link reliability. Temperature variations impact atmospheric conditions, influencing signal propagation. Planners consider these effects to optimize link performance and account for temperature-related changes in their designs. Gaseous absorption, particularly by atmospheric water vapor and oxygen, is factored into link planning by selecting frequency bands less susceptible to absorption, particularly for long-distance links. Lastly, the impact of multipath fading is mitigated through strategies like antenna diversity and adaptive modulation, as well as terrain-aware antenna placement. By navigating these geoclimatic challenges, microwave link planners ensure robust and effective communication links in diverse environmental conditions.

The tool is located in CE RLP tab, Radio Links section.

Run Link Predictions again. After successful calculations, leave MW001 (A-B) link selected, and click on Performance tab.

Calculations

Profile | Interference | Performance | Capacity

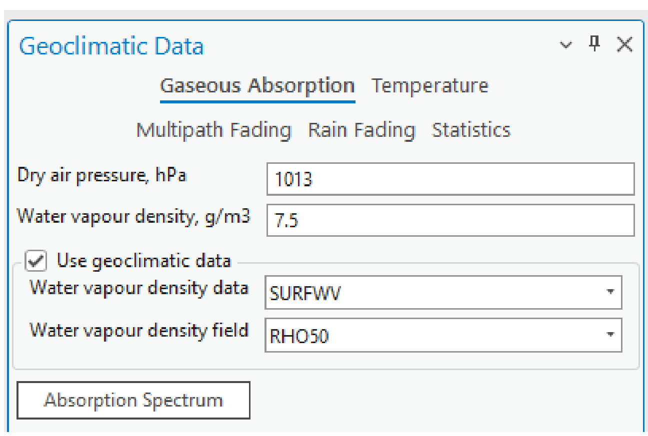
Propagation reliability Total reliability Diversity improvement 1.00 Protection improvement 4166.33

	Total Outage Annual	Total Outage W.Month	Multipath ITU Annual	Multipath ITU W.Month	Multipath V.-Barnett Annual	Multipath V.-Barnett W.Month	Rain ITU Annual	Rain ITU W.Month	Rain Crane Annual	Rain Crane W.Month	Multipath XPD W.Month	Rain XPD W.Month
Unavailability, %	0.00018	0.00584	0.00000	0.00000	0.00000	0.00000	0.00018	0.00073	0.00000	0.00000	0.00511	0.00000
Availability, %	99.99982	99.99416	100.00000	100.00000	100.00000	100.00000	99.99982	99.99927	100.00000	100.00000	99.99489	100.00000
Objective, %	99.99800		99.99900		99.99900		99.99900		99.99900			
Unavailability time	0:00:57	0:30:42	0:00:00	0:00:00	0:00:00	0:00:00	0:00:57	0:03:49	0:00:00	0:00:00	0:26:53	0:00:00
Availability time	8765:59:03	8765:29:18	8766:00:00	8766:00:00	8766:00:00	8766:00:00	8765:59:03	8765:56:11	8766:00:00	8766:00:00	8765:33:07	8766:00:00
Objective time	0:00:00		8765:54:44		8765:54:44		8765:54:44		8765:54:44			

Review Multipath ITU and Rain ITU percentage values.

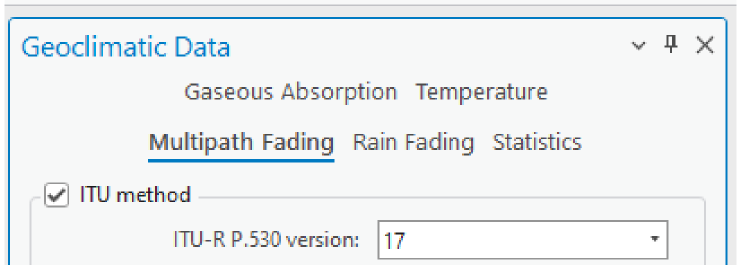
Multipath ITU Annual	Multipath ITU W.Month	Rain ITU Annual	Rain ITU W.Month
0.00000	0.00000	0.00018	0.00073
100.00000	100.00000	99.99982	99.99927
99.99900		99.99900	
0:00:00	0:00:00	0:00:57	0:03:49
8766:00:00	8766:00:00	8765:59:03	8765:56:11
8765:54:44		8765:54:44	

Close results, and open Geoclimatic Data tool.

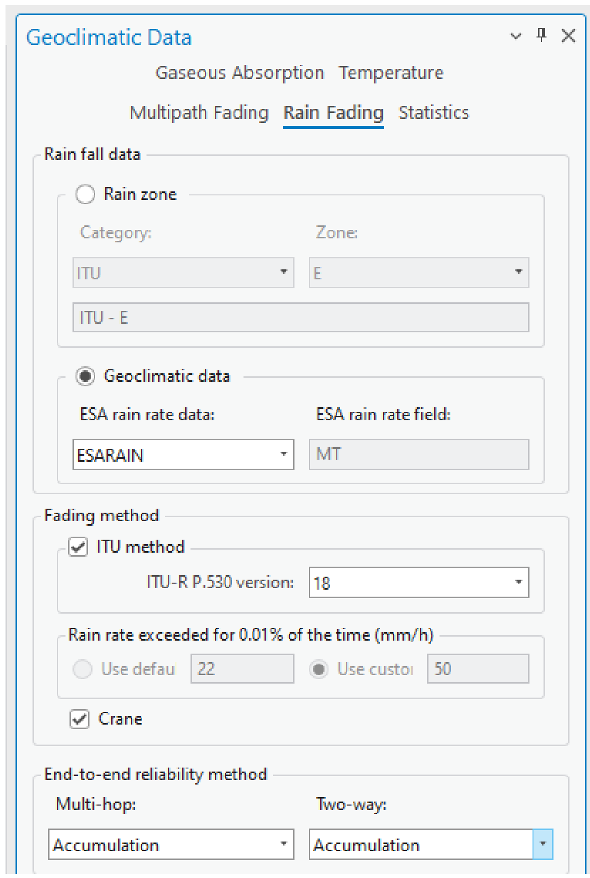


Click on Multipath Fading tab, and define:

- ITU-R P.530 version: 17



Click on Rain Fading tab, and define:



Press Save Changes. Close Geoclimatic Data tab, and run Link Predictions again. Review Performance tab calculations for MW001 (A-B) link. Compare them with previous results.